

Shrub Delineation U-Net — Method & Results Report

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This report documents the full pipeline for the MAMBO shrub-delineation U-Net at Strawberry Hills: data preparation, the architecture change from 5 to 4 max-pooling layers, and a head-to-head comparison of three training runs (baseline, border-weighted loss, and an alternative architecture).

1. Background

The task is binary semantic segmentation of individual shrubs from RGB drone imagery of Strawberry Hills, using an Attention U-Net trained on 512x512 patches extracted from a larger orthomosaic. Shrubs are small, densely packed, and visually similar to background vegetation, so the model's main failure modes are (a) false-positive shrub predictions in non-shrub vegetation and (b) poor delineation between adjacent shrub crowns.

Early runs showed these two failure modes directly traceable to training-data quality and network depth rather than model capacity, which motivated a sequence of interventions: fixing a patch-extraction bug, adding explicit negative (no-shrub) samples, and reducing the network's downsampling depth from 5 to 4 max-pooling layers. This report records what each change did, and what three subsequent training runs (baseline, border-weighted loss, and an alternative architecture) show about where the pipeline stands now.

2. Data Preparation Pipeline

Patch extraction bug. `preprocess.py` slid a fixed-size window across the orthomosaic and its label raster to cut training patches. Near the image edges this window could fall partly outside the real raster extent, producing patches whose image data was black/near-blank but whose label mask still carried real shrub pixels copied from the valid region — a direct contradiction: the network was being told "no visual signal here, but predict shrub anyway." `find_broken_patches.py` scans the existing `mambo_output` (train and test) for exactly this signature (image pixel range below a near-black threshold, non-

empty label) and moves affected pairs to a `discarded_broken_patches` folder rather than deleting them, leaving every other patch — including prior manual cleanup — untouched.

Negative sampling. The original patch set was drawn mostly from shrub-dense areas, giving the model little exposure to genuine "no shrub" vegetation. Adding explicit negative (empty-label) patches gave the loss a clean signal for what background vegetation looks like, directly reducing false-positive shrub predictions on open ground and canopy gaps.

Manual cleanup. On top of the automated broken-patch removal, negatives were manually reviewed and pruned for cases that were technically valid but ambiguous or mislabeled.

Together, the broken-patch fix and negative sampling were the first-order correction: before them, the model was trained on contradictory signal in part of the dataset and had too few genuine negatives to learn a sharp shrub/background boundary from.

3. Architecture Change: 5 to 4 Max-Pooling Layers

The Attention U-Net originally used 5 max-pooling stages in its encoder. At 512x512 input resolution, 5 downsampling steps push the bottleneck feature map down to 16x16 — too coarse to preserve the boundary detail of shrub crowns that are often only a few metres (a handful of pixels) across. Individual shrubs blurred together in the decoder's upsampled output, and the network over-predicted shrub pixels in textured but non-shrub vegetation, since the deepest features no longer carried enough spatial resolution to distinguish them.

Before this change (5 max-pooling layers, no negative sampling), the model trained to a loss around 0.40 and an IoU around 0.50 on the training set — already weak numbers — and validation performance dropped sharply further from there. Qualitatively, the predicted maps frequently misclassified open background grass areas as shrub, the direct signature of the over-prediction problem described above.

Reducing the encoder to 4 max-pooling stages keeps the bottleneck at a coarser downsampling ratio, preserving more spatial detail through the skip connections. Combined with the cleaned negative sampling, this produced a visible qualitative improvement: fewer false shrub predictions in open/background vegetation, and noticeably sharper delineation between adjacent shrub crowns rather than merged blobs.

Run A (Section 4) is the first fully logged run reflecting both this architecture change and the negative-sampling fix, and serves as the baseline for everything that follows.

4. Experimental Runs

All runs use the same test set (same-site held-out patches) and the same separate new-site set (a different location, the real generalization test), threshold 0.45.

What each metric means

Runs below are labeled **A–C** (rather than by internal folder names) for readability.

- **Loss** (combined binary cross-entropy + Dice loss): the quantity the network is directly trained to minimize. Lower is better; it isn't a percentage, so it's only meaningful as a trend between runs, not an absolute score.
- **Accuracy**: the fraction of all pixels classified correctly (shrub or not-shrub). Shrubs cover a minority of each tile, so a model predicting "no shrub" everywhere can still score deceptively high — accuracy alone is not reliable here.
- **Precision**: of the pixels the model labeled as shrub, the fraction that really are shrub. Low precision means too many false shrub predictions (false positives) in open ground or background vegetation.
- **Recall**: of the pixels that really are shrub, the fraction the model found. Low recall means real shrubs are being missed (false negatives).
- **F1 score**: the harmonic mean of precision and recall — one number that penalizes a model for being strong on one and weak on the other. The main metric used to compare runs here.
- **IoU (Intersection over Union)**: predicted-vs-true shrub area overlap divided by their union. Stricter than F1 (penalizes both over- and under-prediction more harshly) and is the standard segmentation metric in remote sensing.

Every metric is reported two ways: **same-site test** (held-out patches from the training location — how well the model fits data like what it trained on) and **new-site** (a separate, unseen location — real-world generalization, the harder and more meaningful test).

Run	Best epoch	Train loss	Test F1	Test IoU	New-site F1	New-site IoU
A — Baseline (4-level Attention U-Net)	16	0.332	0.799	0.670	0.705	0.549
B — Border-weighted loss	18	0.322	0.793	0.662	0.706	0.548
C — ID U-Net (Wagner et al., 2019)	20	0.418	0.791	0.659	0.673	0.514

- **A — Baseline:** the 4-level Attention U-Net architecture (Section 3), trained on the cleaned negative-sampled dataset. Binary (1-channel, sigmoid) output. First fully logged run, used as the reference point for everything below.
- **B — Border-weighted loss:** same architecture and data as A, but the combined BCE+Dice loss is reweighted to penalize errors on pixels near shrub-crown boundaries more heavily, aimed at sharper delineation between adjacent crowns.
- **C — ID U-Net (Wagner et al., 2019):** a 3-channel output variant following the individual-crown-delineation approach of Wagner et al. Instead of one binary mask, the network predicts three classes — FULL shrub shape, a thin BORDER band, and an eroded INNER core (each shrub mask shrunk inward by 2 pixels, $\approx 2\text{cm}$ at this imagery's $\sim 10\text{mm/pixel}$ resolution) — on the idea that shrinking can break the thin "neck" where two touching crowns are fused in the FULL mask, so they come out as two separate blobs in the INNER channel. **Important caveat:** the metrics in the table above are computed on the FULL channel only, which is the same binary shrub/not-shrub signal as A and B — kept comparable on purpose. The actual separation step (connected-component labeling of the INNER channel, which is what would turn touching shrubs into distinct instances) was never run or evaluated in this comparison, so C's instance-separation ability is untested here, not proven. The 2-pixel erosion depth was also not tuned or validated against this imagery's actual touching-shrub geometry — it was chosen as a plausible starting point for the $\sim 10\text{mm/pixel}$ resolution, nothing more.

5. Prediction Overlays

Same tile, same original image, predicted shrub pixels overlaid in orange (threshold 0.45) for each run.

Image — Original image and predicted shrub overlays for the three methods, same tile, t...

A — Baseline; B — Border-weighted loss; C — ID U-Net (Wagner et al., 2019).

At this zoom level the three prediction panels look visually similar — the differences the metrics pick up (Section 4) are fine-grained (border precision, scattered false positives) and hard to see by eye at full-tile scale. The qualitative jump that mattered visually happened earlier and is common to all three panels: moving from 5 to 4 max-pooling layers plus adding negative samples (Sections 2–3) is what took the model from blurred, over-predicted shrub blobs (with grass regions misclassified as shrub) to cleanly delineated

shrub shapes. Note what this image does not show: none of the three panels demonstrates separated individual crowns where shrubs touch or overlap — see the limitation flagged in Section 6.

6. Discussion

The architecture + negative-sampling fix was the single biggest win, and it happened before any of the four logged runs: 5→4 max-pooling plus adding real negative patches took the model from blurry, over-predicted shrub masks to properly delineated individual crowns. That gain isn't visible in the table in Section 4 because every logged run already includes it — it's the baseline everyone is compared against, not a row in the comparison.

Border-weighted loss (B) roughly held the baseline's new-site performance (F1 0.706 vs. 0.705, IoU 0.548 vs. 0.549). It's not an improvement over the baseline, but it didn't regress anything either — the safer of the two alternative configurations tried.

The alternative architecture (C) was worse across the board — higher loss at every stage, worst new-site F1 (0.673) and IoU (0.514). Not worth pursuing in its current form.

Open limitation: overlapping/touching shrub crowns are not currently separated by any evaluated method. A and B are binary semantic segmentation — by design they output one connected "shrub" blob wherever crowns touch, with no notion of individual instances, regardless of resolution or any tiling/margin parameter. C's 3-channel design was meant to solve exactly this (see Section 4), but the connected-component step that would actually separate crowns from its INNER channel was never run or evaluated here, so this remains unverified rather than solved. A separate, independent attempt — watershed splitting directly on the binary prediction mask — was tried outside this comparison and was abandoned as unreliable (too many false splits, slow). Separating overlapping shrubs is therefore still open work, not a solved part of the pipeline.

The real bottleneck is new-site generalization, not same-site fit. Same-site test F1/IoU barely move across the three runs (0.791–0.799 F1, 0.659–0.670 IoU), while new-site metrics swing more (F1 0.673–0.706). Neither border-weighted loss nor the architecture swap meaningfully closed that gap — all three runs plateau around $F1 \approx 0.67$ –0.71 on the new site. The model already fits the training distribution well; what it doesn't do is transfer to genuinely new terrain.

7. Conclusions & Next Steps

1. **Keep the 4-maxpool architecture with negative sampling** — it is the source of the real qualitative gain (sharper delineation, fewer false positives) and underlies every run compared here.
2. **Border-weighted loss (B) is a safe default** going forward — neutral to slightly positive, no downside seen so far — but on its own it isn't the fix for new-site generalization.
3. **Drop C, the ID U-Net variant**, in its current form; it underperforms on every metric.
4. **Make instance separation of overlapping shrubs an explicit, tested deliverable** — this is a hard requirement, not a nice-to-have. Concretely: actually run connected-component labeling on C's INNER channel and measure instance-level precision/recall against hand-labeled touching-shrub pairs (not just the FULL-channel metrics in the table above); re-tune the INNER erosion depth (currently a default 2px $\approx$ 2cm at $\sim$ 10mm/pixel) against real touching-shrub geometry instead of leaving it as an untested starting guess; and treat the abandoned binary-mask watershed (Section 6) as a dead end, not a fallback to retry.
5. **Prioritize new-site generalization directly**, since that's the actual bottleneck (F1 plateaued $\approx$ 0.67–0.71 across all three runs): more training data from additional sites, domain-adaptation or augmentation targeting domain shift (illumination, vegetation composition, seasonality), or evaluating on a third held-out site to see whether the plateau is a property of the new-site data itself rather than the model.

References

Wagner, F.H., Ferreira, M.P., Sanchez, A., Hirye, M.C., Zortea, M., Gloor, E., Phillips, O.L., De Souza Filho, C.R., Shimabukuro, Y.E., Aragão, L.E.O.C. (2019). Individual tree crown delineation in a highly diverse tropical forest using very high-resolution satellite images. *ISPRS Journal of Photogrammetry and Remote Sensing*, 145(Part B), 362–377. <https://doi.org/10.1016/j.isprsjprs.2018.09.013>

Note: often informally cited as "Wagner et al. 2020" since the article appeared online in late 2018 ahead of its dated 2019 print issue; the DOI above resolves to the citable record.